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#include <iostream>
#include <vector>
#include <map>
#include <algorithm>
#include <stdexcept>
#include <string>
using namespace std;

struct User {
    string name;
    int age;
    string greet() const {
        return "Hi, I'm " + name + " (" + to_string(age) + ")";
    }
};

int sum(const vector<int>& nums) {
    int total = 0;
    for (int n : nums) total += n;
    return total;
}

int find_index(const vector<string>& words, const string& needle) {
    for (size_t i = 0; i < words.size(); ++i) {
        string a = words[i], b = needle;
        transform(a.begin(), a.end(), a.begin(), ::tolower);
        transform(b.begin(), b.end(), b.begin(), ::tolower);
        if (a == b) return static_cast<int>(i);
    }
    throw runtime_error("not found: " + needle);
}

int main() {
    User u{"Avi", 33};
    cout << u.greet() << "\n";

    vector<int> nums{1,2,3,4};
    cout << "sum: " << sum(nums) << "\n";

    vector<string> words{"Go","Python","Java"};
    try {
        int idx = find_index(words, "python");
        cout << "found at " << idx << "\n";
    } catch (const exception& e) {
        cout << "error: " << e.what() << "\n";
    }

    map<string,int> m{{"a",1},{"b",2}};
    for (const auto& [k,v] : m) {
        cout << k << " => " << v << "\n";
    }
}

```